DSA5205 Project 1: Machine Learning in Finance

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Abstract

This report studies model complexity, regularization, and out-of-sample performance in return prediction. Task 1 reproduces the misspecified ridge experiment of Kelly, Malamud, and Zhou (2024); the remaining sections document the extensions, hidden-data predictions, and financial interpretation of neural-network generalization.

1 Task 1: Misspecified Simulation with Ridge

1.1 Design and implementation

Following Kelly, Malamud, and Zhou [Kelly et al.(2024)Kelly, Malamud, and Zhou], returns are generated as

$$R_{t+1} = S_t'\beta^* + \varepsilon_{t+1}, \qquad S_t \sim \mathcal{N}(0, I_P), \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{1}$$

The vector β^* is drawn from an isotropic Gaussian distribution and normalized so that $\|\beta^*\|_2^2 = b^* = 0.2$. The data-generating complexity is fixed at $c = P/T_{\text{tr}} = 10$. In each replication, one random permutation of the P columns is held fixed across the full grid. For a given revealed fraction q, only the first $P_1 = qP$ permuted columns are observed, so observed complexity is $cq = P_1/T_{\text{tr}}$. This nested observation rule creates misspecification while preserving the sufficiently mixed setting.

The simulation parameters are listed in Table 1. Data generation is implemented by `generate_data()` in `task1/generate_data.py`; the experiment loop and nested feature selection are in `task1/task1.py`.

Table 1: Task 1 simulation parameters.

Parameter	Value
Base seed and replications	5205; 50
Training and test sizes	$T_{\text{tr}} = 500$; $T_{\text{te}} = 5000$
DGP complexity and dimension	$c = 10$; $P = 5000$
Signal strength and noise	$b^* = 0.2$; $\sigma_\varepsilon = 1$
cq grid	$\{0.50, 0.75, 0.90, 0.95, 0.98, 1.02, 1.05, 1.10, 1.25, 1.50, 2, 3, 5, 7, 10\}$
Ridge grid	$z \in \{0, 0.01, 0.1, 1, 10, 50\}$

For each observed block, ridge is estimated without an intercept:

$$\widehat{\beta}(z) = \left(X'X + zT_{\text{tr}}I\right)^{-1} X'y. \tag{2}$$

At $z = 0$, the implementation uses the Moore–Penrose pseudoinverse. For $z > 0$, it solves the primal system when $P_1 \le T_{\text{tr}}$ and the equivalent dual system otherwise; see `fit_ridge()` in `task1/fit_ridge.py`.

All quantities are evaluated on the independent test set. With $\widehat{R}_{t+1} = S_t'\widehat{\beta}$ and timing return $R^{\pi}_{t+1} = \widehat{R}_{t+1}R_{t+1}$, the reported metrics are

$$R^2_{\text{paper}} = 1 - \frac{\mathbb{E}[(R_{t+1} - \widehat{R}_{t+1})^2]}{\mathbb{E}[R^2_{t+1}]}, \tag{3}$$

$$\mathbb{E}[R^{\pi}_{t+1}] = \mathbb{E}[\widehat{R}_{t+1}R_{t+1}], \tag{4}$$

$$\text{SR} = \frac{\mathbb{E}[R^{\pi}_{t+1}]}{\sqrt{\mathbb{E}[(R^{\pi}_{t+1})^2]}}, \qquad \|\widehat{\beta}(z)\|_2. \tag{5}$$

Their sample analogues are computed by `calculate_metrics()` in `task1/test_metrics.py`.

1.2 Results

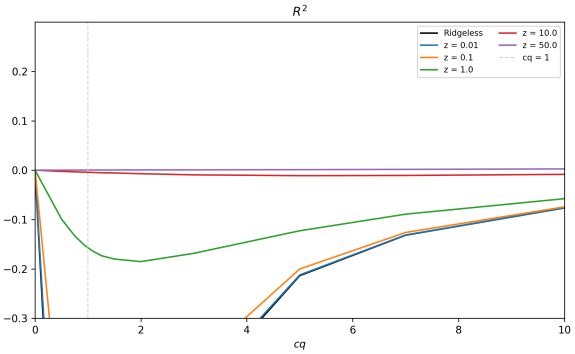
Figure 1 reports averages over the 50 replications. It is the finite-sample analogue of Figures 4–6 in Kelly, Malamud, and Zhou [Kelly et al.(2024)Kelly, Malamud, and Zhou], which plot population limits rather than simulated estimates. The ridgeless specification displays the expected instability around the interpolation threshold $cq = 1$: R^2_{paper} falls to about -50 at $cq = 0.98$–1.02, while $\|\widehat{\beta}\|_2$ rises above 7.5. These extremes, and timing returns above 0.02, are clipped by the displayed axes. As in the paper's Figure 4, both effects abate for $cq > 1$ because overparameterized minimum-norm regression supplies implicit regularization. Explicit shrinkage smooths the interpolation singularity; $z = 10$ or 50 keeps R^2_{paper} close to zero and the coefficient norm small.

The expected-return panel has the same broad message as the left panel of the paper's Figure 5: very simple models have low timing returns because they observe too little of the DGP, and additional features improve approximation. In the population benchmark, ridgeless expected return rises to $b^*/c = 0.02$ at $cq = 1$ and is exactly flat thereafter. The simulation approaches roughly 0.018–0.019 away from interpolation, but overshoots and fluctuates near $cq = 1$; for example, its ridgeless mean is 0.0347 at $cq = 1.02$. This difference is consistent with finite training and test samples and the nonlinear effect of near-singular estimates, rather than a contradiction of the limiting result. The paper's Figure 5 also reports portfolio volatility; the project instead requires the Sharpe ratio and coefficient norm, which expose the same interpolation instability through risk adjustment and parameter dilation.

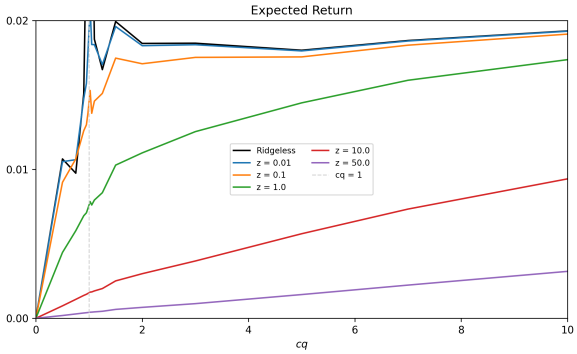
The Sharpe panel closely reproduces the paper's Figure 6. Ridgeless and lightly regularized curves show "double ascent": Sharpe rises at low complexity, dips near interpolation as coefficient and portfolio risk explode, and rises again for $cq > 1$. Nontrivial shrinkage removes most of this dip, producing an approximately permanent ascent. At $cq = 10$, all curves converge to about 0.049–0.050; the largest value on the simulated grid is 0.0505 for $z = 50$. For $\Psi = I$, the paper's optimal shrinkage is

$$z^*(q) = \frac{c\{1 + b^*(1-q)\}}{b^*} = 60 - 10q, \tag{6}$$

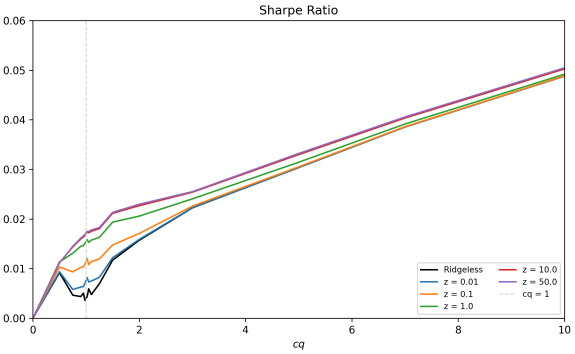
so $z = 50$ is exactly optimal at $q = 1$ and is close to the theoretical range over the grid. Theorem 1's monotone, concave improvement applies to this optimally adjusted $z^*(q)$, not to every fixed-z curve. Overall, the finite-sample experiment supports the virtue-of-complexity mechanism: under misspecification and sufficient mixing, observing more signals improves risk-adjusted timing performance even when conventional out-of-sample R^2 remains negative.



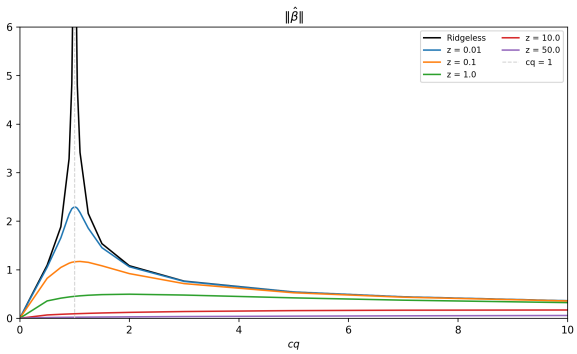
(a) R^2_{paper}.



(b) Expected timing return.



(c) Paper Sharpe ratio.



(d) Coefficient norm.

Figure 1: Finite-sample counterparts of Figures 4–6 in Kelly, Malamud, and Zhou [Kelly et al.(2024)Kelly, Malamud, and Zhou], averaged over 50 replications. The dashed line marks $cq = 1$; displayed axes clip the interpolation extremes. All four panels are produced by `plot_results()` in `task1/plot_results.py` from `results.csv`.

2 Task 2: Alternative Model

2.1 Method and experimental control

2.2 Results and comparison with ridge

3 Task 3: Predictions for Data Sets A, B, and C

3.1 Validation and model selection

3.2 Final models and prediction files

4 Task 4: Generalization and Overfitting in Neural Networks

4.1 Main finding of Montanari and Urbani

4.2 Implication for financial machine learning

5 Task 5: Video and Reproducibility

6 Conclusion

References

[Kelly et al.(2024)Kelly, Malamud, and Zhou] Kelly, B. T., Malamud, S., and Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503.

[Montanari and Urbani(2025)] Montanari, A. and Urbani, P. (2025). Dynamical decoupling of generalization and overfitting in large two-layer networks. *Advances in Neural Information Processing Systems*, 38.